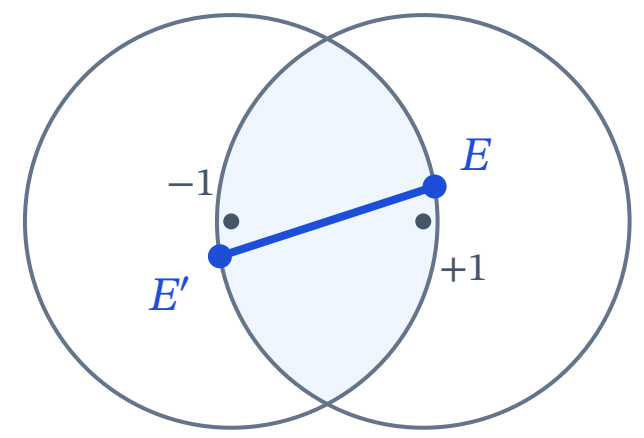


The critical radius of GG_5

$$r_c(5) = \sqrt[3]{3 + \varphi} = 2.1489611417... \quad \text{a root of } x^4 - 7x^2 + 11$$

Theorem. The group generated by the clockwise 72° turns of two closed disks of radius r about ± 1 is finite for $r < \sqrt[3]{3 + \varphi}$ and infinite for $r \geq \sqrt[3]{3 + \varphi}$, where φ is the golden ratio.

The dependency graph of the machine-checked proof in Lean 4 and Mathlib, drawn from its Verso blueprint. Every box is a blueprint node.



The chord $E'E$ in the lens at $r = r_c(5)$

